SUPPORT VECTOR MACHINE – REGRESSION

[2]:
```
import pandas as pd
import pickle
import matplotlib.pyplot as plt
```

[10]:
```
dataset = pd.read_csv('Startups_dataset.csv')
dataset
```

[10]:

	Unnamed: 0	R&D Spend	Administration	Marketing Spend	State	Profit
0	0	165349.30	136897.90	471784.20	New York	192261.93
1	1	162597.80	151377.69	443898.63	California	191792.16
2	2	153441.61	101145.65	407934.64	Florida	191050.49
3	3	144372.51	118671.95	383199.72	New York	182902.09
4	4	142107.44	91391.87	366168.52	Florida	166188.04
5	5	131877.00	99814.81	362861.46	New York	156991.22
6	6	134615.56	147198.97	127716.92	California	156122.61
7	7	130298.23	145530.16	323876.78	Florida	155752.70
8	8	120542.62	148719.05	311613.39	New York	152211.87

[11]:
```
dataset.columns
```

[11]:
```
Index(['Unnamed: 0', 'R&D Spend', 'Administration', 'Marketing Spend', 'State',
       'Profit'],
      dtype='str')
```

[12]:
```
# Apply one-hot encoding
dataset = pd.get_dummies(dataset, columns=['State'], drop_first=True)
dataset
```

[12]:

	Unnamed: 0	R&D Spend	Administration	Marketing Spend	Profit	State_Florida	State_New York
0	0	165349.30	136897.90	471784.20	192261.93	False	True
1	1	162597.80	151377.69	443898.63	191792.16	False	False
2	2	153441.61	101145.65	407934.64	191050.49	True	False
3	3	144372.51	118671.95	383199.72	182902.09	False	True
4	4	142107.44	91391.87	366168.52	166188.04	True	False
5	5	131877.00	99814.81	362861.46	156991.22	False	True
6	6	134615.56	147198.97	127716.92	156122.61	False	False
7	7	130298.23	145530.16	323876.78	155752.70	True	False
8	8	120542.62	148719.05	311613.39	152211.87	False	True

[13]:
```
dataset.columns
```

[13]:
```
Index(['Unnamed: 0', 'R&D Spend', 'Administration', 'Marketing Spend',
       'Profit', 'State_Florida', 'State_New York'],
      dtype='str')
```

[17]:
```
x = dataset[["R&D Spend", "Administration", "Marketing Spend", "State_Florida", "State_New York"]]
y = dataset[["Profit"]]
```

[19]:
```
from sklearn.model_selection import train_test_split
x_train,x_test,y_train,y_test = train_test_split(x,y,test_size=0.20,random_state=24)
```

[20]:
```
x_train
```

[20]:

	R&D Spend	Administration	Marketing Spend	State_Florida	State_New York
20	76253.96	113867.40	298664.57	False	False
33	55494.05	103057.59	214634.91	True	False
27	72107.70	127864.65	353183.91	False	True
30	61994.58	115641.38	91131.34	True	False
10	101913.18	110594.21	229161.05	True	False
49	0.10	116983.90	45173.16	False	False
42	23641.03	96189.73	148001.21	False	False
21	78389.57	153773.53	299737.39	False	True
14	119943.34	156547.52	256513.02	True	False

[22]:
```
from sklearn.preprocessing import StandardScaler
sc = StandardScaler()
x_train=sc.fit_transform(x_train)
x_test=sc.transform(x_test)
```

[126]:
```
from sklearn.svm import SVR
reg = SVR(kernel = "linear", gamma = 'auto', C = 1800)
reg.fit(x_train,y_train)
```

```
C:\Users\Admin\anaconda3\Lib\site-packages\sklearn\utils\validation.py:1365: DataConversionWarn
expected. Please change the shape of y to (n_samples, ), for example using ravel().
  y = column_or_1d(y, warn=True)
```

[126]:

SVR

▸ Parameters

▸ Fitted attributes

[127]:
```
y_pred = reg.predict(x_test)
y_pred
```

[127]:
```
array([150014.63811637, 157274.68801453,  92786.54763134, 107338.44261103,
       106817.13921787, 170614.81323636, 117382.7042733 ,  81690.28458401,
       102907.84495074,  87990.39918913])
```

[128]:
```
from sklearn.metrics import r2_score
score = r2_score(y_test,y_pred)
score
```

[128]:
```
0.9272830923533087
```